Assignment 3 : : Total Marks: 50

Fourier Series

1. Determine the Fourier series of the periodic function $f(x)$ defined as follows:

$$f(x) = \begin{cases} x, & 0 \leq x \leq \pi, \\ 2\pi - x, & \pi < x \leq 2\pi. \end{cases}$$

   The function is periodic with period 2π.

2. Find the Fourier series expansion of the function $f(x) = \sin(2x) + \cos(3x)$ on the interval $-\pi \leq x \leq \pi$.

3. Find the Fourier series of the following piecewise function:

$$f(x) = \begin{cases} 1, & -\pi \leq x \leq 0, \\ -1, & 0 < x \leq \pi. \end{cases}$$

   The function is periodic with period 2π.

4. Compute the Fourier series of the function $f(x) = x^2$ on the interval $-\pi \leq x \leq \pi$.

5. Calculate the Fourier series for the function $f(x) = e^x$ defined on the interval $0 \leq x \leq 2\pi$.

Fourier Transform

1. Compute the Fourier transform of the function $f(t) = e^{-t^2}$, which is a Gaussian function.

2. Find the Fourier transform of $f(t) = \cos(2t)$.

3. Determine the inverse Fourier transform of the function $F(\omega) = \frac{1}{1+\omega^2}$.

4. Calculate the Fourier transform of $f(t) = \text{rect}(t)$, where $\text{rect}(t)$ is the rectangular function.

Orthogonal Functions

1. Verify whether the functions $f_1(x) = \sin(x)$ and $f_2(x) = \cos(x)$ are orthogonal on the interval $[0, \pi]$. That is, check if

$$\int_0^{\pi} f_1(x) f_2(x)\, dx = 0.$$